LAB 11: CHATBOT SYSTEM INTEGRATION

Course: Programming for Artificial Intelligence | Instructor: Rasikh Ali

1. Lab 11 Scope & Objectives

Lab 11 focuses on the **deployment and humanization** of intelligent systems. While previous labs covered the theoretical foundations of Artificial Neural Networks (ANNs), Lab 11 bridges the gap between a standalone AI model and a user-centric web application.

Core Focus: Full-stack integration of a Neural Network backend with a Flask-based frontend interface.

2. Task 11: Intelligent Chatbot Workflow

The primary task involves creating a seamless communication channel between a user and an AI agent. The chosen implementation is a **Mental Health Support Bot**, designed to provide empathetic interaction.

A. The Data Layer (Intent Mapping)

In Lab 11, we structure "Intents" which act as the knowledge base for the ANN. This includes:

- **Patterns:** Sample user queries (e.g., "I'm feeling stressed").
- **Tags:** Classification labels the model learns to recognize.
- **Responses:** Humanized outputs mapped to specific tags.

B. The Logic Layer (Flask & NLP)

The Python backend serves as the bridge. It performs three critical operations:

- **Text Preprocessing:** Using NLTK for Tokenization and Stemming to normalize user input into a format the model understands.
- **Inference:** Feeding the processed vector into the Keras model to predict the highest probability intent.
- **Response Generation:** Randomly selecting a response from the predicted category to ensure the conversation feels natural.

3. Advanced Programming Concepts

Lab 11 utilizes several high-level programming paradigms:

- AJAX/jQuery **Asynchronous Communication:** Ensuring the web page does not reload when the user sends a message, providing a smooth "app-like" experience.
- Flask Routing **RESTful API Design:** Implementing `POST` methods to securely handle user data and return JSON-formatted bot replies.
- Dense Neural Networks **Vector Classification:** Leveraging a Sequential ANN architecture to map sparse "Bag of Words" vectors to intent classes.

4. Humanized Interface Design

A key requirement of Lab 11 is "Humanization." This is achieved through:

- **Dynamic Chat UI:** A responsive interface that mimics modern messaging apps (WhatsApp/Slack style).
- **Empathetic Scripting:** Carefully crafted responses that provide validation and support, moving beyond technical data retrieval.
- **Error Handling:** Providing fallback responses when the model's confidence threshold is low.